

Technical Procedure Manual: Geotechnical Investigation, Foundations, and Reinforced Concrete Structures

1. Geotechnical Investigation: Standard Penetration Test (SPT) Protocols

Strategic Context

The Standard Penetration Test (SPT) is not merely a routine survey; it is the primary risk-management tool in civil engineering. The N -value derived from this process is the technical and legal basis for calculating the allowable soil pressure (q_{adm}), which dictates the entire structural design. As Field Operations Director, I consider any deviation from ABNT sampling protocols a breach of professional liability. Inaccurate geotechnical data leads to structural pathology, unmanageable settlement, or catastrophic collapse.

Apparatus and Standards: Pre-Survey Compliance Checklist

Immediate field suspension is mandatory if equipment does not meet the following ABNT specifications. Verify and initial each item before drilling:

- **Hammer:** Mass of 65 kg (iron, cylindrical/prismatic) with a coxim of hard wood.
- **Drop Height:** Precisely 75 cm free fall, verified by a guide mark on the rod.
- **Sampler:** External diameter of 50.8 mm and internal diameter of 34.9 mm; must be replaced if the bico (shoe) is worn.
- **Rods:** Internal diameter of 25 mm, straight, with clean threads. Total mass must be approx. 3.23 kg/m.
- **Borehole Density Plan:** Minimum of 2 holes for areas up to 200 m²; 3 holes for 200–400 m². For projects between 1,200 m² and 2,400 m², add one hole for every additional 400 m².

Operational Execution: Drilling and Soil Classification

- **Initial Phase:** Utilize a *trado-concha* (100 mm) to a depth of 1.0 m.
- **Dry Advancement:** Continue with a helical auger (*trado helicoidal*) until the water table is reached.
- **Washing Transition (*Lavagem*):** Shift to water circulation using a *trépano* (washing bit) if penetration is < 50 mm after 10 minutes of dry drilling or if soil adherence prevents auger rotation.
- **Classification:** Field logs must use precise terminology:
- **Sand:** Classify by texture (Fine: 0.05–0.42 mm; Medium: 0.42–2.0 mm; Coarse: 2.0–4.8 mm).
- **Clay:** Classify consistency from "Very Soft" (flows through fingers) to "Hard" (cannot be molded, loses structure under pressure).
- **Qualifiers:** Use "Variegated" (*variegado*) for multi-colored layers and "Micaceous" where mica is present.

Sampling and Dynamic Penetration (N-Values)

Record the number of blows for each 15 cm interval of a 45 cm drive. The N -value is the sum of the final 30 cm.

- **Refusal:** If penetration is null after 5 successive hammer impacts, the test is stopped and marked "Impenetrable."

- **Accountability:** If refusal occurs above 8 m depth, the engineer must relocate the sounding up to four times within 2 m of the initial point to confirm a localized obstruction versus a bearing stratum.

Field Reporting and Hydrostatic Monitoring

- **Water Table (WT):** Record the exact time of the initial reading and a secondary reading at 24 hours.
- **Casing Position:** The depth of the *revestimento* (lining tube) at the time of WT reading is mandatory for accurate hydrostatic pressure assessment. **Structural Connection:** These results provide the q_{adm} . If the N -value is misrecorded, the foundation sizing will be flawed, creating a direct risk of settlement (*recalque*) that compromises the superstructure's integrity.

2. Site Logistics, Infrastructure, and Quality Management Systems

Strategic Context

Site organization is a direct indicator of project profitability. A chaotic "Canteiro de Obras" erodes margins through material degradation and rework. Adherence to NBR ISO 9001:2008 and SiAC is the only path to "Total Satisfaction" and the elimination of *Desperdício* (Waste).

Living Areas and Amenities: Regulatory Ratios

The following ratios are the legal minimum. Failure to maintain these results in immediate site citations: | Facility | Requirement Ratio | Technical Specification || ----- | ----- | ----- || **Toilet/Vaso Sanitário** | 1 per 20 workers | Minimum area 1.0 m²; must be sifonados. || **Lavatory** | 1 per 20 workers | Height of 90 cm; 60 cm spacing between taps. || **Shower** | 1 per 10 workers | 0.80 m² min; individual stalls; mandatory hot water. || **Dining Area** | Total Workforce | Tables with smooth, washable tops; must be independent of toilets. || **Bebedouros** | 1 per 25 workers | Filtered, fresh water with inclined jet nozzles. |

Storage and Warehouse Management

The **Storekeeper (Almoxarife)** is the gatekeeper of the project's liquid assets.

- **Layout:** Materials must be stored at least 2 m from the site perimeter to prevent theft.
- **Inventory:** Critical levels (sand, cement, f_{ck} additives) must trigger automatic alerts to prevent work stoppage.

The Quality Framework (ISO 9001/SiAC)

- **Process Management:** Control the inputs and processing, not just the output.
- **Continuous Improvement:** Every non-conformity must trigger a root-cause analysis to prevent recurrence. Waste mitigation is a daily operational objective.

3. Execution and Control of Foundation Systems

Strategic Context

The foundation is the irreversible interface between the structure and the earth. As Director, I demand rigorous technological control for deep foundations, where visual inspection is impossible after casting.

Deep Foundations: Methodology and Procedure

- **Strauss Piles:** Cast-in-place; ensure the bucket and probe are free of debris to prevent soil contamination of the concrete.
- **Franki Piles:** High load capacity through an expanded base (*base alargada*).
Procedure: The base must be formed by driving a "plug" of dry concrete into the soil with high-energy blows until the required volume is achieved.
- **Continuous Flight Auger (CFA):** Must be **electronically monitored** . The operator must provide a real-time digital log of torque, drilling pressure, and concrete injection volume to ensure no necking or discontinuities in the shaft.

Technical Controls: Nega and Cutoff

- **Nega (Refusal):** Confirm the bearing stratum by measuring the permanent set (penetration) over the **last 10 blows** of the hammer. This value must be recorded and compared against the design assumptions.
- **Cota de Arrasamento (Cutoff Level):** The final trim elevation. If the pile is trimmed too low or left with contaminated "cold joints," the capping beam connection will fail.
- **Tolerances:** Any eccentricity or inclination deviation beyond ABNT limits requires an immediate structural recalculation before the capping beam (*Bloco de Coroamento*) is cast.

4. Reinforced Concrete Structures: Materials and Preparatory Stages

Strategic Context

Structural longevity is non-negotiable. Synchronization of formwork, rebar, and concrete quality is the only way to prevent premature structural deflection and cracking.

Aggregates and Cement Control

- **Cement:** Store in dry, covered areas. **Command:** Stacking must not exceed **10 bags in height** to prevent "caking" (hydration due to pressure). Any caked cement must be rejected.
- **Aggregates:** Sand must be free of organic impurities and classified as Fine, Medium, or Coarse per the source terminology.

Steel Reinforcement (Armadura)

Pre-Pour Mandatory Inspection:

- **Cleanliness:** Bars must be free of grease, oil, paint, or loose rust.
- **Positioning:** Verify the concrete cover (*cobrimento*) using spacers.
- **Rigidity:** Ensure the cage is tied to prevent displacement during vibration.

Formwork and Stripping (Desforma)

- **Materials:** Selection between wood and metal depends on reuse cycles.
- **Stripping Timing:** Premature stripping is a leading cause of permanent structural deflection. Removal must follow a strict sequence based on the concrete's developed strength, not the project schedule.

5. Concrete Pouring, Admixtures, and Technological Control

Strategic Context

Concrete pouring is an irreversible chemical event. Real-time "Controle Tecnológico" is the only assurance against structural failure.

Mixing and Workability

- **Slump Test:** Mandatory for every truck to verify workability before discharge.
- **Additives:**
- **Plasticizers/Superplasticizers:** Use to increase flow without adding water (which reduces f_{ck}).
- **Retarders:** Mandatory for large pours or long transport times.
- **Accelerators:** Used for rapid formwork turnover.

Pouring and Vibration (Adensamento)

- **Drop Limit:** Concrete fall must not exceed 2 meters to prevent segregation.
- **Vibration Technique:**
- **Mechanical Vibration:** The vibrator must be inserted vertically. Respect the "radius of action" to ensure overlap of influenced areas.
- **Risk:** Avoid touching the rebar; do not over-vibrate, as this causes **internal segregation** , where aggregates sink and grout rises.
- **Manual Vibration:** Only permitted for small volumes where mechanical means are inaccessible; requires "energetic" manual tamping.

Curing and Laboratory Analysis

- **Cura (Curing):** Surfaces must remain moist for a minimum of **7 days** to prevent plastic shrinkage.
- **f_{ck} Analysis:** Field-molded cylindrical specimens are the legal proof of the project's safety. If the f_{ck} results fall below the characteristic design strength, the structure must be extracted and analyzed via core samples (*testemunhos*) or reinforced at the contractor's expense.**Final Compliance:** Documentation of every hammer blow and every cubic meter of concrete is the field engineer's shield against professional liability and the only guarantee of a high-integrity build.